

Ratio Notation and Sharing

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the ratio 15 : 25 in its simplest form.

1 The highest common factor of 15 and 25 is 5.

2 $15 \div 5 = 3$ and $25 \div 5 = 5$

Answer: 3 : 5

2 [2 marks]

Write the ratio 24 : 36 : 60 in its simplest form.

1 Look for a number that divides into all three. 12 works.

2 $24 \div 12 = 2$, $36 \div 12 = 3$, $60 \div 12 = 5$

3 2, 3 and 5 share no further common factor, so this is fully simplified.

Answer: 2 : 3 : 5

3 [2 marks]

Share £45 in the ratio 4 : 5

1 $4 + 5 = 9$ parts altogether.

2 One part = $45 \div 9 = £5$

3 4 parts = £20 and 5 parts = £25 (check: $20 + 25 = 45$ ✓)

Answer: £20 and £25

4 [2 marks]

Write the ratio 2 : 5 in the form 1 : n

1 Divide both sides by 2 so the left-hand side becomes 1.

2 $2 \div 2 = 1$ and $5 \div 2 = 2.5$

Answer: 1 : 2.5

5 [3 marks]

120 sweets are shared between three children in the ratio 3 : 4 : 5

How many sweets does each child get?

1 $3 + 4 + 5 = 12$ parts.

2 One part = $120 \div 12 = 10$ sweets.

3 3 parts = 30, 4 parts = 40, 5 parts = 50

4 Check: $30 + 40 + 50 = 120$ ✓

Answer: 30, 40 and 50 sweets

6

[3 marks]

In a box of counters the ratio of red to blue is 2 : 7

There are 14 red counters.

(a) How many blue counters are there?

(b) How many counters are there altogether?

1 2 parts are worth 14 counters, so 1 part = 7 counters.

2 (a) Blue = 7 parts = $7 \times 7 = 49$ counters

3 (b) Total = $2 + 7 = 9$ parts = $9 \times 7 = 63$ counters

4 (Or simply $14 + 49 = 63$.)

Answer: (a) 49 (b) 63

7

[3 marks]

Write the ratio 45 minutes : 2 hours in its simplest form.

1 Both quantities must be in the same units first.

2 2 hours = 120 minutes, so the ratio is 45 : 120

3 The highest common factor of 45 and 120 is 15.

4 $45 \div 15 = 3$ and $120 \div 15 = 8$

Answer: 3 : 8

8

[3 marks]

A pastry mix uses flour and butter in the ratio 5 : 2

350 g of flour is used.

(a) How much butter is needed?

(b) What is the total mass of the mixture?

1 5 parts = 350 g, so 1 part = $350 \div 5 = 70$ g

2 (a) Butter = 2 parts = $2 \times 70 = 140$ g

3 (b) Total = $350 + 140 = 490$ g

Answer: (a) 140 g (b) 490 g

9

[4 marks]

Some money is shared in the ratio 3 : 8

The larger share is £75 more than the smaller share.

Work out the total amount of money.

1 The difference between the parts is $8 - 3 = 5$ parts.

2 Those 5 parts are worth £75.

3 1 part = $75 \div 5 = £15$

4 The total is $3 + 8 = 11$ parts, so $11 \times 15 = £165$

5 Check: shares are £45 and £120, and $120 - 45 = £75$ ✓

Answer: £165

10

[4 marks]

$a : b = 2 : 3$ and $b : c = 4 : 5$

Find the ratio $a : b : c$ in its simplest form.

- 1 The letter b appears in both ratios, so make its value match.
- 2 b is 3 in the first ratio and 4 in the second. The lowest common multiple of 3 and 4 is 12.
- 3 Multiply the first ratio by 4: $a : b = 8 : 12$
- 4 Multiply the second ratio by 3: $b : c = 12 : 15$
- 5 Now b agrees, so $a : b : c = 8 : 12 : 15$
- 6 8, 12 and 15 share no common factor, so this is simplest form.

Answer: 8 : 12 : 15

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Write the ratio $0.4 : 1.2$ in its simplest form.

- 1 Multiply both sides by 10 to remove the decimals: $4 : 12$
- 2 Divide both sides by 4.

Answer: 1 : 3

2

[2 marks]

Write the ratio $5 : 8$ in the form $n : 1$

- 1 This time divide both sides by 8 so the right-hand side becomes 1.
- 2 $5 \div 8 = 0.625$ and $8 \div 8 = 1$

Answer: 0.625 : 1

3

[3 marks]

$x : y = 3 : 4$ and $y : z = 6 : 7$

Find $x : y : z$

- 1 y is 4 in the first ratio and 6 in the second; the lowest common multiple is 12.
- 2 Multiply the first ratio by 3: $x : y = 9 : 12$
- 3 Multiply the second ratio by 2: $y : z = 12 : 14$
- 4 Combining gives $9 : 12 : 14$

Answer: 9 : 12 : 14

4

[3 marks]

In a class the ratio of boys to girls is 5 : 7

(a) What fraction of the class are boys?

(b) There are 21 girls. How many students are in the class?

1

(a) There are $5 + 7 = 12$ parts, of which 5 are boys, so the fraction is $\frac{5}{12}$

2 (b) 7 parts = 21 girls, so 1 part = 3 students.

3 Total = 12 parts = $12 \times 3 = 36$ students

Answer: (a) $\frac{5}{12}$ (b) 36 students

5

[4 marks]

A bag contains red and blue counters in the ratio 3 : 5

12 more red counters are added and the ratio becomes 9 : 10

Work out the total number of counters now in the bag.

1 Let the original numbers be $3x$ red and $5x$ blue.

2 After adding: $(3x + 12) : 5x = 9 : 10$

3 Cross-multiply: $10(3x + 12) = 9 \times 5x$, so $30x + 120 = 45x$

4 $15x = 120$, giving $x = 8$

5 Red is now $3(8) + 12 = 36$ and blue is $5(8) = 40$

6 Total = 76 (check: $36 : 40$ simplifies to $9 : 10$ ✓)

Answer: 76 counters

6

[4 marks]

A money box contains 20p coins and 50p coins in the ratio 5 : 3

Six of the 50p coins are removed, and the ratio becomes 5 : 2

How many coins were in the box originally?

1 Let the original numbers be $5x$ twenty-pence coins and $3x$ fifty-pence coins.

2 After removing six: $5x : (3x - 6) = 5 : 2$

3 Cross-multiply: $2 \times 5x = 5(3x - 6)$, so $10x = 15x - 30$

4 $5x = 30$, giving $x = 6$

5 Originally there were $5(6) = 30$ twenty-pence and $3(6) = 18$ fifty-pence coins.

6 Total = 48 coins (check: $30 : 18$ simplifies to $5 : 3$ ✓)

Answer: 48 coins

7

[4 marks]

A bag contains 10p, 20p and 50p coins in the ratio 4 : 5 : 6

The total value of the coins is £8.80

How many coins are there altogether?

- 1 Let the numbers be $4x$, $5x$ and $6x$. Work in pence.
- 2 Value = $4x \times 10 + 5x \times 20 + 6x \times 50$
- 3 = $40x + 100x + 300x = 440x$ pence
- 4 £8.80 = 880p, so $440x = 880$ and $x = 2$
- 5 The numbers of coins are 8, 10 and 12, giving 30 coins in total.
- 6 Check: $80p + 200p + 600p = 880p$ ✓

Answer: 30 coins

8

[4 marks]

Some money is shared between A, B and C in the ratio 2 : 3 : 7

C receives £120 more than B.

Work out how much each person receives.

- 1 The difference between C and B is $7 - 3 = 4$ parts.
- 2 4 parts = £120, so 1 part = £30
- 3 A = $2 \times 30 = £60$
- 4 B = $3 \times 30 = £90$ and C = $7 \times 30 = £210$
- 5 Check: $210 - 90 = £120$ ✓ (total £360)

Answer: A £60, B £90, C £210

9

[4 marks]

Paint is mixed using white and blue in the ratio 7 : 3

Tom needs 25 litres of the mixture but only has 6 litres of blue paint.

(a) Show that Tom cannot make 25 litres of the mixture.

(b) Work out the greatest volume of mixture he can make.

- 1 (a) Blue is 3 parts out of $7 + 3 = 10$, so it makes up $\frac{3}{10}$ of the mixture.
- 2 $\frac{3}{10}$ of 25 = 7.5 litres of blue paint is needed.
- 3 Tom only has 6 litres, so he cannot make 25 litres.
- 4 (b) With 6 litres of blue, 3 parts = 6, so 1 part = 2 litres.
- 5 The mixture is 10 parts = $10 \times 2 = 20$ litres.

Answer: (a) 7.5 litres of blue is needed (b) 20 litres

A map has a scale of 1 : 50 000

A rectangular field measures 4 cm by 3 cm on the map.

(a) Work out the real dimensions of the field in kilometres.

(b) Work out the real area of the field in km^2 .

(c) Explain why the area on the map is not $\frac{1}{50\,000}$ of the real area.

1 (a) 1 cm on the map represents 50 000 cm = 500 m = 0.5 km.

2 4 cm represents $4 \times 0.5 = 2$ km and 3 cm represents $3 \times 0.5 = 1.5$ km

3 (b) Real area = $2 \times 1.5 = 3 \text{ km}^2$

4 (c) Lengths are scaled by 50 000, but area involves two lengths multiplied together.

5 So areas are scaled by $50\,000^2 = 2\,500\,000\,000$, i.e. 2.5×10^9

6 (Check: the map area is 12 cm^2 and $3 \text{ km}^2 = 3 \times 10^{10} \text{ cm}^2$, and $3 \times 10^{10} \div 12 = 2.5 \times 10^9 \checkmark$)

Answer: (a) 2 km by 1.5 km (b) 3 km^2 (c) areas scale by the square of the length scale factor, $50\,000^2$